

Novogene Microbial FAQ



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1.1.1 How to interpret the distribution plot of the base content of second-generation sequencing data.

The base content distribution plot demonstrates the proportion of GC distribution at different positions of reads with varying lengths. Different colored curves represent the proportions of A, T, G, C, and N. Due to the use of random PCR amplification and paired-end sequencing, the proportions between AT and GC should generally be consistent. However, at the beginning of the reads, there may be some fluctuations due to preferential effects caused by primer ligation.

1.1.2 Is there a significant difference between the GC content of clean data and the SequenceGC content?

First of all, GC content refers to the specific GC content in the genome of a species or in a specific DNA or RNA fragment. This situation usually occurs in contaminated samples. CleandataGC% represents the proportion of GC at each position of the reads and is calculated based on all reads. SequenceGC% represents the proportion of GC in the assembled sequence. If the sample is not contaminated and the PE reads sequencing is random, both values should be equal. However, if the sample is contaminated, the calculation of CleandataGC% includes the contaminated reads, while the assembled sequence is more complex and may contain contaminated sequences, making the proportion uncertain. Therefore, the two values may not be equal.

1.1.3 What are the filtering conditions for converting raw data from second-generation sequencing to clean data?

The readfq.v10 software is used for quality control. The data quality received by the teacher has already been filtered using the software to remove low-quality bases. The fourth line in the fq file of each read represents the quality of each base. The filtering conditions are as follows:

Filter out reads with an N ratio higher than 10%.

Filter out reads with a proportion of base quality values below 20 higher than 40%.

Filter out duplications (completely identical paired-end reads).

Filter out paired-end reads containing sequencing adapters (15 bases aligned to adapter sequences with a maximum of 3 mismatches).

1.1.4 Why are the lengths of Read 1 and Read 2 in second-generation sequencing data not consistent?

Due to technical limitations in sequencing, the quality of nucleotides at the end of the reads may decrease, especially for samples with higher GC content. The decrease in quality can be more pronounced. To ensure the quality of assembly, we trim and remove the low-quality regions at the end of reads with an average quality below a certain threshold, usually selecting a Q value of 10 or below, corresponding to an average error rate of 10%. Generally, Read 1 has better quality than Read 2, so Read 2 is trimmed more. Therefore, after trimming, Read 1 is usually longer than Read 2.

1.1.5 What is Q-value, and what are Q20 and Q30?

Q-value is the result of evaluating the quality score obtained by rounding the conversion of raw sequencing data into bases during Illumina's base sequencing process. In this process, the quality score (-10lg e) is calculated based on the error rate (e). For example, a Q-value of 20 corresponds to an approximate error rate of 0.01, while a Q-value of 30 corresponds to an error rate of 0.001. Q20 and Q30, on the other hand, represent the proportions of bases in reads with Q-values higher than 20 or 30, respectively, reflecting the overall quality of the sequencing.

1.1.6 How are the data in the Errorrate graph calculated?

The Errorrate we calculate is the expected value of the error rate at individual base positions, using a logarithmic average. The simple calculation method is to average the Q-values of that base position and convert it back to an error rate. For example, if the average Q-value at a certain position in the reads is 20, the corresponding error rate would be 0.01.

1.2 Survey

1.2.1 What is K-mer and what is its function?

K-mer refers to a DNA sequence of length K, where K is a positive integer. For example, if K=17, it is called a 17-mer. K-mer serves various purposes such as error correction in sequencing, contig assembly, and estimation of genome size, heterozygosity rate, and repetitive sequence content. If the length of a sequencing read is L, then there can be L-K+1 K-mers in a read.

When estimating genome size using K-mers, the formula is: Genome size = Total number of K-mers / Expected K-mer coverage depth.

For microbial genomes, choosing a K-mer size of 15 is often sufficient. This is because there are four different bases (ATCG), and there are 4^{15} possible nucleotide fragments of length 15, which is enough to cover the size of microbial genomes. However, larger K-mers are not necessarily better due to the presence of erroneous bases in reads. Choosing a larger K-mer size would result in more K-mers containing the erroneous position.

1.2.2 Why can genome size estimation be inaccurate?

The estimation of genome size can be inaccurate due to the mathematical models we use, which are based on shotgun sequencing and random sampling. These models require a sequencing coverage depth that is sufficiently deep (generally 50X or higher, with a minimum of 30X) to approximate the assumptions of the mathematical model. In cases where the coverage depth is insufficient, the estimated genome size may have a bias due to the inaccurate assumptions of the model. Additionally, contamination from exogenous sequences can also lead to deviations in genome size estimation.

1.2.3 Which situations in a survey indicate the occurrence of contamination?

Typical indicators of contamination in a survey include: the absence of peaks in the k-mer plot, the presence of multiple peaks in the k-mer plot, and an estimated genome size significantly larger than expected. When these situations occur, there is a high possibility of external DNA contamination, which needs to be further confirmed by subsequent preliminary assembly, GC-depth analysis, and NT library alignment.

1.2.4 Why is it necessary to conduct a survey before performing a fine mapping of fungal genomes?

The genomes of fungi are quite complex, which is why it is necessary to conduct a survey beforehand. The survey helps in determining whether further steps, such as adding large fragment libraries for assembly, are needed. The primary evaluation criteria include heterozygosity and sample contamination. Heterozygosity has a significant impact on genome assembly, and it is generally required to be below 0.5%.

1.2.5 In Kmer graph, the presence of small peaks does not necessarily indicate sample contamination.

When the genome size is relatively small, if the shotgun sequencing technique is explained using a sampling model, both the sample population and the number of sampling times are

significantly lower than the ideal values. As a result, the results will deviate noticeably from a normal statistical distribution (such as the Poisson distribution). Additionally, due to the fluctuation in frequencies, the curve of the peaks may not be smooth, but this is a normal phenomenon and not indicative of sample contamination. Those small peaks are not caused by sample contamination and fall within the normal range. Please rest assured. These small fluctuations are due to the smaller genome size and are not related to contamination.

1.2.6 The meaning of Heterozygous Rate in the all_Kmer.stat.xls file

In the all_Kmer.stat.xls file, the Heterozygous Rate refers to the estimated proportion of heterozygosity in the genome. Organisms with high heterozygous rates are more difficult to assemble, and it is generally recommended that the heterozygous rate should not exceed 0.5%. A heterozygous rate of "0" indicates that the strain does not possess any heterozygosity.

1.3 Assembly and evaluation

1.3.1 What are N50 and N90 in the assembly results?

N50 and N90 are commonly used assembly metrics in genome assembly. They represent the lengths of sequences in the assembly results. To calculate N50 (N90), the sequences are sorted in descending order by length, and the total length of sequences that are longer than a specific length is calculated. N50 (N90) is defined as the length of the sequence where the total length of sequences is just greater than 50% (90%) of the genome length. This value reflects that more than 50% (90%) of the genome is covered by sequences of that length or longer, indicating the quality contribution of the assembly for subsequent analysis.

1.3.2 How is the GC-depth plot generated? What is its significance?

The GC-depth plot represents the relationship between the GC content and sequencing depth distribution across the entire genome. The specific method involves dividing the genome into segments of a certain length, where each segment has a specific GC content and reads coverage depth, corresponding to a point in the plot.

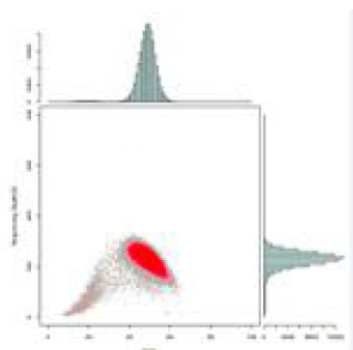
For a specific sample with relatively pure DNA, the GC content and depth will be concentrated in a certain range and disperse towards the surrounding regions. The further away from the central range, the fewer sample points will be found. If the GC-depth plot separates into multiple distinct clusters, it generally indicates that the assembled genome contains DNA from different sources, particularly if there is a clear separation in the GC

Sometimes, when the GC content does not separate but only the depth does, it could be an indication of DNA originating from plasmids. In such cases, it is necessary to combine other information, such as the results of nucleotide alignment, for further analysis.

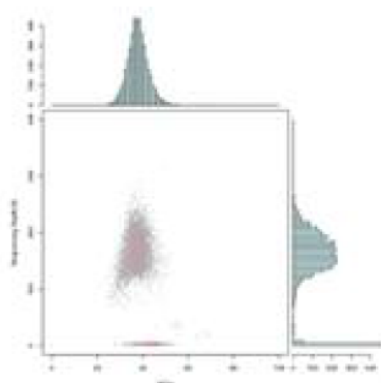
In summary, the GC-depth plot provides valuable insights into the composition and potential contamination of the assembled genome.

1.3.3 What are the possible scenarios and reasons for multiple centers in the analysis and statistical plot correlating GC content with sequencing depth (Depth)?

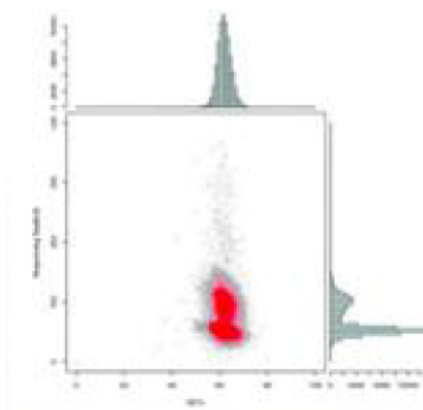
- 1) It is possible for fungi to have mitochondria, which is normal.



- 2) In case of slight contamination with miscellaneous fungi, isolation can be attempted.



- 3) In case of contamination with closely related species, it is more difficult to achieve complete isolation.



1.3.4 Why can't we achieve good assembly results in the presence of contamination and mixing?

Due to the assumption made by assembly software that the sequencing data originates from a single genome, the presence of exogenous DNA contamination introduces sequences with varying degrees of similarity and dissimilarity from different sources. These complex relationships can disrupt the assembly software, which, in order to ensure accuracy, can only fragment suspicious regions into separate sequence fragments. As a result, the final assembly only provides fragmented sequences, losing the intended effect of assembly. If a closely related reference genome can be identified for contaminant separation, it is possible to improve the aforementioned results to some extent. However, due to the potential similarity between exogenous DNA and the target genome, as well as potential differences between the target and reference genomes, there is a risk of both false positives and false negatives in the separation process. Therefore, regardless of the approach, it is impossible to achieve the standard of pure DNA in the assembly after separation. Due to the limitations imposed by contamination and the reference genome, we cannot guarantee successful assembly results for such samples.

1.3.5 Difference between sequencing coverage and sequencing depth

(1) Sequencing depth refers to the ratio of the total number of sequenced bases to the size of the target genome. For example, if a gene has a size of 2M and the sequencing depth is 10X, the total amount of data obtained would be 20M. Coverage refers to the proportion of sequenced reads that align to the entire genome. Due to the presence of complex structures such as high GC content and repetitive sequences in the genome, the sequences obtained from sequencing and assembly often do not cover all regions, resulting in gaps. For example, if the coverage of a bacterial genome sequencing is 98%, it means that 2% of the sequence regions were not obtained through sequencing.

(2) Sequencing coverage and sequencing depth are two different concepts. We ensure that the data volume is sufficient, guaranteeing a sequencing depth, which is the ratio of

the sequencing data volume to the predicted genome size at 100X. This is shown as 100X in the report.

1.3.6 Is there any gap in the assembly, and if so, how was it filled?

For unfinished projects, the current assembly result may have gaps as the assembly has not been completed yet. Specific gap information can be found in the result file 02.Assembly/NO/NO.gapInfo.xls. The file 02.Assembly/Assembly.readme.doc provides an explanation of the 02.Assembly/NO/NO.gapInfo.xls file.

To achieve a gapless assembly or a complete genome, one can perform additional sequencing with larger insert sizes to help connect scaffolds. Primers can be designed on both sides of the gaps, and Sanger sequencing can be used to sequence the sequences within the gaps for gap filling. However, this process is time-consuming and requires dedicated manpower.

1.3.7 How to handle contamination in the project?

Usually, we will conduct contamination assessment and provide feedback on whether the assembly results can be effectively separated. In cases where effective separation is possible and approved by the customer, we can attempt separation and annotation as part of the after-sales service. However, we cannot guarantee the accuracy and completeness of sequence separation.

1.3.8 How to find plasmid information for sequencing strains?

Customers can use the blast tool on NCBI to perform nt database alignment, and obtain plasmid information from the alignment results.

1.3.9 Why do some completed library samples form plasmid circles while others do not? How are plasmids confirmed?

We discovered the sequencing depth of chromosome reads to be around 100x, while the sequencing depth of circular plasmids is around 80x, and that of linear plasmids is only around 20-40x. Therefore, it is likely that these samples have a lower copy number of plasmids, possibly due to their instability and potential loss, resulting in insufficient sequencing depth to assemble the plasmids into circular forms.

For plasmid confirmation, we first align these shorter sequences with all genome sequences. If the aligned sequences do not exceed 50% of their own length, they are retained. Sequences exceeding 50% are excluded, and the remaining shorter sequences are submitted to NCBI to check if they align with plasmid sequences. If they align with

If the customer requires a guarantee of circular plasmid delivery, this can be discussed during the early stages. We can ensure plasmid assembly into circular forms and delivery by increasing the sequencing data volume.

1.3.10 Bacterial finishing graph assembly process (PacBio)

Genome assembly using SMRTLink v5.0.1 software:

The raw data obtained from sequencing contains a certain proportion of low-quality reads. To ensure accurate results in subsequent information analysis, the raw data needs to be filtered (removing reads below 500bp) to obtain clean data.

The SMRTportal software's automatic error correction function is used to select long reads (above 6000bp) as seed sequences. The shorter reads are aligned with the seed sequences using Blasr to improve the accuracy of the seed sequences. This process results in an initial assembly that represents the genome.

Correction of the initial assembly:

The variantCaller module in SMRTLink software is used to correct and analyze the variant positions in the initial assembly using the arrow algorithm.

Correction of third-generation data using second-generation data:

The corrected assembly from step 2 is used as the reference genome. The second-generation reads are aligned to the assembly using the bwa software, and the alignment results are filtered based on criteria such as a minimum base quality of 20, a minimum read depth of 4, and a maximum read depth of 1000. The corresponding positions in the initial assembly are then corrected based on the filtering results.

Circularization and start point correction:

The corrected assembly from step 3 is self-aligned to check for overlaps at the ends. Based on the presence of an overlap of more than 1.5k, it can be determined whether the chromosomal sequence forms a circular genome. The assembly is then compared to DNA databases to correct the start points. Subsequent alignment analysis is conducted against plasmid databases to distinguish between chromosomal and plasmid sequences, resulting in the final assembled genome.

The criterion for determining whether the chromosomal sequence forms a circular genome based on overlaps is as follows: A 2K segment is extracted from the beginning of the assembled sequence and compared to a 60K segment from the end. If an overlap of more than 1.5k is detected, it is considered a circular genome.

1.3.11 Bacterial finishing graph assembly process (Nanopore)

Based on the quality-controlled and valid data from each sample, the genome assembly of second-generation and third-generation sequencing data is performed using the Unicycler software.

Initial assembly: Firstly, the second-generation data is assembled into a draft assembly using SPAdes software, which generates a contig graph.

Bridging: Then, the third-generation data is aligned to the contig graph using miniasm and Racon software. The long reads from the third-generation data are used to bridge gaps in the contig graph.

Bridging application: When conflicts arise, Unicycler selects the highest-quality bridging based on the corresponding quality scores.

Circularization determination and final confirmation:

- a. The ends of the contigs are checked for overlaps to determine if circularization has occurred.
- b. Unicycler uses default markers (such as dnaA or repA) and start_genes to correct the start points and confirm the final assembly.
- c. The assembly is then filtered to distinguish between chromosomal and plasmid sequences, and the chromosomal sequence is assembled into a circular genome (or a linear genome if it is a linear genome sequence). This results in the final assembly of the genome without any gaps (0gap).

Therefore, the final output is the complete circular genome sequence.

1.3.12 Fungal polished assembly process (PacBio)

1、Initial assembly:

Use SMRTLink v5.0.1 software for genome assembly:

The raw data obtained from sequencing contains a certain proportion of low-quality reads. To ensure the accuracy of subsequent analysis results, the raw data needs to be filtered (removing reads shorter than 500bp) to obtain Clean data.

Use the automatic error correction function in SMRTportal software to select longer reads (greater than 6000bp) as seed sequences. The shorter reads are aligned to the seed sequences using Blasr to further improve the accuracy of the seed sequences. The assembly result obtained provides an initial reflection of the genome.

2、Correction of the initial assembly result:

Use the variantCaller module in SMRTLink software to correct and count the variant sites in the initial assembly result using the arrow algorithm.

1.4 Genome assembly and functional annotation

1.4.1 Can GeneMark be used for bacterial gene annotation?

The Genemark package includes a series of software, among which GenemarkS is a version that contains a self-training module. It is capable of creating training models based on the sequence itself, and it has built-in gene models for various organisms including bacteria and viruses. However, for eukaryotes that contain introns, the models are more complex, and there is another software called GenemarkES specifically designed for gene annotation in eukaryotes with introns. GenemarkS is currently recognized as one of the software with good annotation performance for bacteria.

1.4.2 What are the three methods for fungal gene prediction (de novo prediction, homology-based prediction, and transcriptome-based prediction)?

To predict genes, the Augustus software can be used for ab initio prediction, while the Genewise software can be used for homology-based prediction. Homology-based prediction requires providing coding gene cDNA sequences or protein sequence information from the same species. It is important to note that the closer the homologous species, the better the prediction results, and information from non-related species is not very informative and can lead to inferior results. Therefore, for the fungal genome project, it is best if the customer can provide coding gene information (cDNA or protein) from closely related species or assembled transcriptome sequence files. If the teacher cannot provide or does not have coding gene information or transcriptome information from closely related species, then gene prediction can only be performed based on the previously used ab initio prediction algorithm.

The results obtained from these three methods will be integrated using the EVM software. Therefore, it would be ideal if the teacher can provide reference sequences (already gene annotated) from closely related species and transcript data, and combine the prediction results from the three methods.

Transcriptome data prediction / Homology-based prediction / De novo prediction (gene prediction): Any of these can be used for analysis. If two or three types of data are provided, the intersection of the results from the three methods or two methods can be taken. However, the proportions of the results may vary, with transcriptome > homology-based > ab initio prediction. Generally, predicted protein sequences start with "Met," but if the alignment results do not start with "M," it is necessary to first check if the provided transcript data has degradation or if the reference genome sequences all start with "M," as both can

affect the alignment results (de novo prediction is not affected). Adjusting the proportions between the three methods or two methods can be attempted to improve the results.

1.4.3 If a gene of interest is not annotated, what could be the reason?

This gene has not been assembled.

This gene does not exist in the target genome.

This gene cannot be annotated because it is not present in the annotation database.

This gene is not present in the strain of the tested organism.

The database version is outdated.

1.4.4 Why can't 5S/16S/23S sequences be annotated for ncRNA?

The above situations occur more frequently when using de novo methods to predict ncRNA sequences. Predicting ncRNA using de novo methods requires complete ncRNA sequences to confirm their structures. However, ncRNAs, especially 16S and 23S sequences, often contain repetitive elements, making it easy to assemble them incompletely. If the entire rRNA sequence is not assembled into a complete sequence, it is not possible to predict the corresponding rRNA sequence. Even if the assembly is good, if the species corresponding to the sample is poorly annotated in the database, it may not be annotated even if there are fewer assembled scaffolds. This does not mean that there is no rRNA present.

In some projects involving newly discovered eukaryotic species, there may be instances where the count of 18S sequences is 0. This is because the 18S sequence of that species has not been sequenced before, so it is not included in the databases and commonly used software, making it impossible to predict the presence of 18S in the assembly results.

1.4.5 Why are some genes not found in GeneBank?

Each sequencing strain will have its own unique genes. Generally, for a bacterium, the conservation gene ratio is around 95%. Therefore, it is normal that some genes cannot be retrieved from the GeneBank database. When performing homology alignment of coding genes, protein-to-protein comparison (blastp) is often used. This is because different bacterial strains have differences in their coding gene sequences, and differences in codon preferences can also affect the similarity of the alignment. If the differences are significant, it may not be possible to align using blastn. Targeted homology alignment using the amino acid sequence of genes can be used for retrieval, which can result in more accurate retrieval of related genes.

1.4.6 How can a specific functional gene be found in the KEGG annotation results?

Method 1: Search for keywords in the file Result/04.Genome_Function/AnnoSummary/***.AnnoSummary.xls. However, this method is relatively vague.

Method 2: Use the annotation results from the KEGG database to search using EC enzyme numbers. For example, if you are looking for cyclohexylamine oxidase, you can search for the keyword "cyclohexylamineoxidase" on the KEGG database website (<http://www.kegg.jp/>), and it will give you the corresponding EC number, which is 1.4.3.12. Then, search for this EC number in the annotation results from KEGG to obtain more accurate results.

Method 3: Provide the nucleotide or protein sequence of the enzyme gene and perform a blast comparison.

1.4.7 What is the difference between EC numbers and KO numbers when matching against the KEGG database?

KEGG ORTHOLOGY is a database of orthologous genes, represented by a format of K followed by a 5-digit number. Each KO represents a group of orthologous genes. For example, K01623 (http://www.kegg.jp/dbget-bin/www_bget?ko:K01623). If a KO represents an enzyme, it often has a corresponding enzyme commission (EC) number, such as EC:4.1.2.13. However, one EC number may correspond to multiple KO numbers. Therefore, KO-based gene annotation provides a more comprehensive count of annotated genes.

1.4.8 What is the difference between Identity, Evalue, and Score in functional annotation results?

Identity refers to similarity, specifically the consistency of a sequence. A higher value indicates a higher homology and greater sequence similarity, suggesting a higher likelihood of performing the same function. There is no fixed range for this value, but based on experience, an identity value above 80% is considered highly reliable, while below 30% has little reference significance.

E-value represents the reliability of the alignment result and is a statistical P-value used to determine the reliability of the alignment. Generally, an E-value below $1e-5$ is considered reliable. E-value and identity are unrelated and cannot be converted.



The identity in an alignment reflects the similarity of the aligned regions. The higher the value, the higher the similarity. However, the identity value does not reflect the overall length of the alignment. On the other hand, the E-value is calculated in combination with the sequence length and represents the expected number of random alignments that would produce a similar result. A smaller E-value indicates that such a scenario is less likely to happen by chance, suggesting a higher probability of a true similar sequence. Therefore, in annotation results, the reliability can be initially judged based on the E-value, with a lower E-value indicating higher reliability. When selecting the best alignment result, the score is often used, and the alignment with the highest score is considered the most similar and reliable.

Sometimes, the identity may exceed 80%, but the E-value displays 0. In reality, this E-value is not exactly 0 but extremely low, rounded to 0. To determine which result is more reliable, simply look at the score. The alignment with the highest score is the most similar and reliable. The score has no defined threshold, with a higher score being better. The score is calculated using a scoring matrix and represents the degree of match between your sequence and the target sequence.

1.4.9 Why can't some genes be found when inputting KO numbers into the KEGG website for analysis?

Some of the genes annotated in the KEGG database are involved in metabolic networks or have metabolic pathway maps that can be found in the KEGG pathway database. However, there are also some genes that do not participate in the metabolic pathway network or do not have metabolic pathway maps in the existing KEGG pathway database. These genes can only be found in the KEGG gene database and cannot be found in the pathway database.

1.4.10 What do the boxed areas and different colors represent in KEGG pathway maps?

The annotated enzymes in the sample are represented by colored boxes, with different colors indicating the number of genes annotated in the sample. The EC number is displayed inside the box, and hovering over the box will display relevant information. Clicking on the box will redirect to the KEGG website for detailed information. Additionally, the corresponding genes annotated to each map number can be found in the "~.kegg.catalog.map.gene.xls" file. The gene_ID can be used to search for the corresponding sequence and information in the "Gene" folder within the "03.Genome_Component" directory. The legend for the colors will be displayed in the result file.

1.4.11 How can the name of an enzyme be determined from the KEGG map code?

Method 1: On the KEGG homepage (<http://www.kegg.jp/>), enter the enzyme code, such as 5.1.1.3, in the search bar and click the search button.

Method 2: The names of the map images we provide are consistent with the map numbers in the KEGG database, such as map00010.png. On the KEGG homepage (<http://www.kegg.jp/>), enter "map00010" to search for this map image, and then you can click on all the boxes to view the corresponding enzyme information.

1.4.12 Some pathways are annotated in the organism but not found in the map.ko table. Why is that?

The reason for this situation is that map351, 472, 901, 5150, and other maps contain KOs that are shared with other maps. In other words, a KO can belong to multiple pathways, so the different KOs are counted in other maps. You can check the file *.kegg.catalog.map.gene.xls in the folder /Result/04.Genome_Function/General_Gene_Annotation/KEGG in the result files. Search for the corresponding map number and see which genes/KOs are present in that map.

1.4.13 How can specific genes, such as the mcr-1 gene, be filtered in KEGG pathway maps?

First, go to the directory Result/04.Genome_Function/AnnoSummary/--*.gbk and find the [gbk format files]. Gbk format files are a type of file that integrates the predicted gene's location, ID, nucleotide sequence, amino acid sequence, and annotated functional information. The format is similar to the gbk files downloaded from NCBI.

The "gene" field corresponds to the gene ID and the "product" field corresponds to the gene name. You can first find the corresponding gene ID based on the gene name.

Next, in the Result/04.Genome_Function/KEGG/--*.kegg.anno.xls file, which contains the annotation results from the KEGG database, search for the corresponding KO_id based on the gene ID.

Then, in the Result/04.Genome_Function/KEGG/*.kegg.catalog.map.gene/ directory, which contains the gene information annotated on the KEGG metabolic pathway maps, find the corresponding map number (KEGG pathway map number) based on the KO_id. Each map number corresponds to a signaling pathway map.

Finally, in the Result/04.Genome_Function/KEGG/--.kegg.map/ directory, which stores the

KEGG functional metabolic pathway maps, check the signaling pathway map based on the map number. The "" corresponds to the pathway number, and each pathway corresponds to a file. The number of genes is indicated by different colors. The file "color_direction.gif" is the corresponding legend.

1.4.14 Where can the specific information of each annotated gene be obtained?

CDS sequences: The CDS sequences are located in the file 03.Genome_Component/Gene/XXX.gmhmp.cds, which is a fasta file that can be opened with a text editor like EditPlus. Each sequence is preceded by a ">" symbol, followed by the gene name. The format after the gene name is "locus=Scaffold1:102:971", which indicates the Scaffold name, Scaffold start position, and Scaffold end position corresponding to the gene.

GFF file: The file 03.Genome_Component/Gene/XXX.gmhmp.gff is a GFF3 format file containing predictions for the genes. It can be opened with a text editor or can be renamed with the extension ".xls" and opened with Excel. The columns from left to right represent the following information:

seqid: Sequence identifier

source: Source of the annotation information

type: Type of annotation information

start: Start position

end: End position

score: Score indicating the likelihood of the annotation information. It can be a numeric value like E-values in sequence similarity alignment or P-values in gene prediction. "." indicates no value.

strand: Strand direction of the sequence. "+" for the sense strand, "-" for the antisense strand, "?" for unknown.

phase: Only valid for annotations of type "CDS". It indicates the coding start position and can take values of 0, 1, or 2.

attributes: This field consists of multiple key-value pairs describing the annotation information. The keys and values are separated by "=" and different key-value pairs are separated by ";". A key can have multiple values, which are separated by ",". Note that if the description includes tab keys and "=", they need to be escaped using URL escape rules. For example, tab key can be represented as "%09". The keys are case-sensitive and keys starting with uppercase letters are predefined and may be referenced by other annotation information later. For more details, refer to the document "03.Genome_Component/Gene/Gene.readme.doc".

1.4.15 How can the amino acid and nucleotide sequences of the genes of interest be found from the result files?

To find the gene IDs corresponding to the relevant functions in the ".kegg.anno" file, you can search for the gene IDs in the first column of the file. Once you have the gene IDs, you can then retrieve the corresponding amino acid sequences and nucleotide sequences from the ".gmhmmp.pep" and "*.gmhmmp.cds" files, respectively.

1.4.16 How is the starting point determined in a complete genome?

We predict the origin of replication by using GC_skew and aligning the upstream sequence of the DnaA gene.

1.4.17 What information can be provided from methylation modifications in a complete genome?

The double-stranded structure of DNA encodes the core information in the form of four different nucleotides: adenine (A), thymine (T), guanine (G), and cytosine (C). However, in biological organisms, DNA bases are often not purely ATGC. The bases A and C can frequently undergo methylation, which can affect the overall methylation level of the genome and subsequently impact the virulence traits of the organism.

Due to the methylation modification, the chemical kinetics of base pairing with the modified bases can differ. When the substrate bases are methylated, the sequencing signals detected during sequencing will be altered. Third-generation sequencing can detect two major modifications involved in bacterial methylation: N6-methyladenine (6mA) and 4-methylcytosine (4mC).

Methylation at different positions plays important regulatory roles during different stages of cell growth. DNA methylation can influence the chemical kinetics of base pairing, thereby affecting gene transcription and expression, reflecting changes in the epigenomic landscape. Our provided results consist of two parts: the type (6mA or 4mC) and location of the methylation modification sites, as well as the type of motif sequence surrounding the methylation sites.

1.4.18 Why are PKS (polyketide synthase) and NRPS (non-ribosomal peptide synthetase) structures not present in the annotation analysis of secondary metabolite gene clusters?

During the analysis, it is performed in two steps. Firstly, we predict the presence of PKS

(Polyketide Synthase) and NRPS (Nonribosomal Peptide Synthetase). Secondly, based on the structure of gene clusters in the current software training set, we make predictions if there are matching structures in the training set. The absence of PKS and NRPS may be due to either the samples themselves not having these two enzymes or the results of these enzymes not matching those in the training set.

1.4.19 Workflow for secondary metabolite analysis

The software used for secondary metabolite analysis is antiSMASH (2.0.2). Its online website is <http://antismash.secondarymetabolites.org/>. The parameter "--clusterblast" is used to BLAST identified clusters against known clusters, and you can submit a gbk file for analysis.

1.4.20 Why can't the KOID be found in the anno file for annotated pathways?

KO refers to KEGG Orthology Group, which represents a group of genes that have the same function. In a KO, there are several genes that perform the same function. Therefore, these gene clusters that perform the same function are called orthology groups. A KOID can be involved in multiple pathways. If a KO cannot be assigned to a specific pathway, it cannot be annotated to that pathway, which is normal.

1.4.21 Why are the starting amino acids not Methionine in protein sequences and why are these proteins truncated?

The software used for predicting secretory proteins is SignalP. Mature secretory proteins are protein sequences that have had their signal peptides (consisting of 15-30 amino acids at the N-terminus of the secretory protein) cleaved. Therefore, the starting amino acid observed in mature secretory proteins is not Met, and the protein sequence is shorter compared to the original protein.

1.4.22 Methods for identification and annotation of P450 genes and the CAZy database

Taking CAZy as an example, the identification method involves using the BLAST software to compare the amino acid sequences of the target species with the CAZy database (Carbohydrate-Active enZymes Database). The gene of the target species and its corresponding functional annotation information are combined to obtain the annotation results. Since each sequence may have multiple alignment results, in order to maintain its biological significance, only the best alignment result is retained as the annotation result.

1.4.23 How to interpret the same abundance of different EC number IDs in the CAZy functional gene abundance table

The identical abundance of these EC numbers in the results is mainly due to the identical sequences predicted for these ECs. As there are multiple EC numbers within the same family that have identical sequences, annotating the same gene sequence against the database will yield multiple EC number results. Therefore, these EC numbers actually represent the alignment results of the same sequence in the database, resulting in the same abundance.

1.4.24 Detailed explanation of the CDS file

CDS stands for Coding Sequence, which refers to the predicted coding sequences obtained from the sequence file using gene prediction software. The CDS sequence file provided in the results displays the starting and ending positions of each CDS, as well as whether it is in the forward or reverse orientation in relation to the original sequence file. The "-" symbol indicates that the CDS is the reverse complement of the corresponding sequence in the sequence file, while the remaining sequences are located in the non-coding regions. In the case of bacteria, the CDS sequences are predicted using the GeneMarkS software (<http://topaz.gatech.edu/GeneMark/>), and the results are automatically generated by the software.

1.4.25 Prophage

The results are stored in the Prophage.stat.xls file. If there is no output, it means that no prophages were predicted. In bacteria, different samples can have different outcomes, and it is normal to not predict any prophages in certain cases.

1.4.26 What is an online prediction website for predicting genomic islands?

Genomic	Island	Prediction	Website:
http://www.pathogenomics.sfu.ca/islandviewer/upload/			

1.4.27 Detailed explanation of the fungal fine circular graphs in the display.

Circle 1: The outermost circle displays the genomic sequence position coordinates.

Circle 2: Genomic GC content. The GC content is calculated using a window size of

200,000bp and a step size of 200,000bp. The blue region towards the center represents regions with GC content lower than the average GC content of the entire genome, while the purple region towards the outer side represents regions with higher GC content. The peak height indicates the difference in GC content from the average.

Circle 3: Genomic GC skew. The GC skew is calculated using a window size of 200,000bp and a step size of 200,000bp. The specific algorithm used is $(G-C)/(G+C)$. The green region towards the center represents regions with lower G content compared to C content, while the pink region towards the outer side represents regions with higher G content compared to C content.

Circle 4: Gene density (coding genes).

Circle 5: Gene density (rRNA).

Circle 6: Gene density (snRNA).

Circle 7: Gene density (tRNA).

Circles 4-7: Gene density of coding genes, rRNA, snRNA, and tRNA respectively. The gene density is calculated using a window size of 200,000bp and a step size of 200,000bp. Darker colors indicate higher gene density within the window.

Innermost circle: Chromosomal duplications.

1.5 Related to result files

1.5.1 How to open the data

In the corresponding folder, there is usually a detailed file description. The result files in bioinformatics are generally divided into two categories. One is a plain text file (or compressed text file) that can be opened with a text editor (or after decompression, open with a text editor). Recommended text editors are Notepad2, <http://www.flossfreeware.ch/notepad2.html>.

The other category is image files, which can be opened with an image viewer, such as the Windows image preview tool. One type of image file to note is the SVG file, which is also an image file but in a vector graphics format for the web. It can be opened with IE 8.0 or above, or third-party browsers like Firefox or Chrome. Older versions of IE may require installation of plugins to display SVG files. It is recommended to use Firefox, as it is more convenient to use.

For commonly used files such as PDF, they can usually be opened with corresponding tools, without further elaboration. Additionally, it is worth mentioning the opening of large files, such as fastq files of cleandata. These files are also plain text files and can be opened with a text editor. However, due to the memory mechanism for reading text files in Windows, the entire file will be loaded into memory, which can cause memory congestion and hinder the opening process. In fact, these files contain the raw data of sequencing, and the specific results are already covered in the result files we provide. Therefore, it is recommended that teachers do not attempt to open similar files in Windows. If there is a need, we can provide a portion of the file content as an example for teachers to understand the content and structure of the file.

1.5.2 How to open and interpret M8 files?

You can open M8 files directly with a text editor (such as EditPlus) or with Excel by changing the file extension to .xls. The M8 file format represents a list of comparison results, where each column from left to right indicates the following: query name, subject name, identity, alignment length, mismatch count, gap count, query alignment start coordinate, query alignment end coordinate, subject alignment start coordinate, subject alignment end coordinate, expectation value, and alignment score. The query refers to the predicted gene, while the subject refers to the gene in the database.

1.6 Advanced analytics

1.6.1 The three methods for constructing a phylogenetic tree

SNP-based tree construction: This method uses SNP matrices of samples and reference strains to build a phylogenetic tree. For each strain, all SNPs are connected in the same order to obtain sequences in fasta format with the same length (one of which is a reference sequence), which are used as input files. The PHYML algorithm of TreeBeST is used to construct the phylogenetic tree based on maximum likelihood.

Core-pan analysis-based tree construction: This method identifies single-copy core genes of samples through core-pan analysis. MUSCLE software is used to align protein sequences and convert them into CDS results. The Treebest's maximum likelihood model is then employed to construct the phylogenetic tree.

Gene family-based tree construction: This method identifies single-copy orthologous genes through gene family clustering. MUSCLE software is used to align protein sequences and convert them into CDS results. The Treebest's maximum likelihood model is used to construct the phylogenetic tree.

1.6.2 Explanation of evolutionary tree branch credibility

The numbers on the branches represent bootstrap values, which are used to assess the credibility of the evolutionary tree branches you have calculated. Simply put, the sequences are shuffled, and the shuffled sequences are then used to construct a tree using the same method. If the branch that was present in the original tree also appears in the tree constructed from the shuffled sequences, one point is assigned to that branch. If it does not appear, zero points are assigned. After repeating this process a given number of times (e.g. 1000 repetitions), each branch is assigned a score, which is then converted into a bootstrap value by the computer.

When constructing a tree for closely related species, the software has higher accuracy in its calculations. However, when including outgroups or distantly related species, the software needs to consider more factors, leading to decreased accuracy and lower bootstrap values. The bootstrap values of evolutionary trees with and without outgroups precisely validate this point.

Explanation of Bootstrap Values:

There is no specific threshold for bootstrap values.

Bootstrap values do not indicate the level of sequence homology. Higher bootstrap values do not necessarily mean higher levels of homology. Bootstrap analysis is a resampling method, also known as a bootstrap method, which involves sampling with replacement. The numerical values in the branches represent the statistical results of bootstrap analysis. For example, if the clustering of sequences A and B occurs in all 1000 bootstrap resampling iterations, the value would be 1000. Converting this to a percentage, it would be 100. If the clustering occurs only in 200 out of 1000 iterations, the value would be 200, equivalent to 20%. This is the principle behind the bootstrap support rate. In cases where the homology is high but the bootstrap value is low, it is because the relationship between sequences A, B, and C is very close, with minimal differences. Whether A and B cluster together or cluster with C, the software considers it to be valid and statistically significant, without being able to distinguish their true phylogenetic relationships. Thus, in a resampling process with shuffling, particularly when many sequences have very low levels of diversity, the chances of A and B actually clustering together in each iteration are quite low. Branch support indicates the reliability of the topology but does not indicate the degree of relatedness. The degree of relatedness is assessed by the branch lengths.

1.6.3 Why are the petal plot/Venn diagram inconsistent with the statistical numbers in the table for shared and unique gene analysis?

The petal plot/Venn diagram represents each sample as an ellipse, and the data in each region represents the number of groups that appear exclusively in that specific region's samples. A group refers to a set of genes with greater than 50% similarity and sequence length differences lower than 0.7. The statistical numbers in the table represent the count

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of individual genes, while the diagram illustrates the count of gene groups.